

CSS Basics Notes

1. What is CSS?

CSS (Cascading Style Sheets) is the language we use to style and design web pages. If HTML is the skeleton (structure) of a webpage, CSS is the skin, clothes, and decorations that make it look nice.

- **Cascading** → Means that rules are applied in a certain order. If two styles conflict, the browser decides which one has higher priority.
- **Style Sheets** → A sheet (or file) full of style rules that can be applied to many HTML pages.

👉 **Real-Life Example:** Think of an online resume: HTML gives the basic text like "Education, Skills, Projects." CSS decides the font, colors, spacing, and how attractive it looks.

2. Why Do We Use CSS?

- **To make websites beautiful:** Add colors, fonts, backgrounds.
- **To control layout:** Arrange text, images, and sections.
- **To save time:** One CSS file can style multiple HTML pages.
- **For responsive design:** Adjusts look for mobile, tablet, and desktop.
- **For better user experience:** Helps people enjoy using the website.

Without CSS, every website would look like plain black-and-white text.

3. Types of CSS

3.1 Inline CSS

Written directly inside an HTML tag using the `style` attribute.

```
<p style="color: red; font-size: 20px;">This is inline styled text.</p>
```

✅ Quick and easy for one element. ❌ Becomes messy in big projects.

3.2 Internal CSS

Written inside `<style>` tags in the `<head>` of an HTML file.

```
<head>
  <style>
    h1 {
      color: blue;
      text-align: center;
    }
  </style>
</head>
```

✅ Useful for single-page projects. ❌ Cannot be reused across multiple pages.

3.3 External CSS

Written in a separate `.css` file and linked to HTML.

```
<!-- In HTML -->
<link rel="stylesheet" href="styles.css">
```

```
/* In styles.css */
p {
  color: green;
  font-size: 18px;
}
```

✅ Best method for large projects. ✅ Reusable and clean.

👉 **Real-Life Example:** Think of a school dress code. One rulebook (external CSS) applies to all students (HTML pages).

4. How to Link External CSS

Use `<link>` tag inside `<head>`:

```
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

- `rel="stylesheet"` → Tells browser it's a stylesheet.
 - `href` → Path of CSS file.
-

5. How CSS Works: Selector and Rules

Syntax:

```
selector {  
  property: value;  
}
```

👉 Example:

```
p {  
  color: red;  
  font-size: 20px;  
}
```

- `p` → selector (chooses `<p>` elements).
 - `color`, `font-size` → properties.
 - `red`, `20px` → values.
-

6. Selecting Elements in CSS

6.1 Universal Selector

Applies to everything.

```
* {  
  margin: 0;  
  padding: 0;  
}
```

6.2 Tag Selector

Applies to specific HTML tags.

```
h1 {  
  color: blue;  
}
```

6.3 Class Selector

Applies to elements with a `class` attribute.

```
.highlight {  
  color: green;  
  font-weight: bold;  
}  
<p class="highlight">This is highlighted text.</p>
```

6.4 ID Selector

Applies to element with unique `id`.

```
#special {  
  color: red;  
}  
<p id="special">This is special text.</p>
```

👉 Rule of Thumb:

- Use **class** when you want to reuse styles.
 - Use **id** only for unique elements.
-

7. CSS Text and Color Properties

7.1 `color`

Sets text color.

```
p {  
  color: purple;  
}
```

7.2 `background-color`

Sets background.

```
div {  
  background-color: yellow;  
}
```

7.3 font-size

Controls text size.

```
h1 {  
  font-size: 40px;  
}
```

7.4 font-family

Defines the font.

```
p {  
  font-family: Arial, sans-serif;  
}
```

👉 Always use a **fallback font** like `sans-serif` in case the first font isn't available.

7.5 font-weight

Controls boldness.

```
h2 {  
  font-weight: bold;  
}
```

7.6 font-style

Italic or normal text.

```
p {  
  font-style: italic;  
}
```

7.7 text-align

Aligns text.

```
h1 {  
  text-align: center;  
}
```

7.8 text-transform

Controls text case.

```
p {  
  text-transform: uppercase;  
}
```

👉 "hello world" → "HELLO WORLD"

8. Working with Fonts

8.1 Google Fonts

- Go to [Google Fonts](https://fonts.google.com/).
- Choose a font and copy the link tag.

```
<link href="https://fonts.googleapis.com/css2?family=Poppins&display=swap" rel="stylesheet">
```

- Apply in CSS:

```
body {  
  font-family: 'Poppins', sans-serif;  
}
```

8.2 Download Fonts from a Website

- Right-click → Inspect → Computed Styles → check font-family.
- Download from sites like **dafont.com** or **fontsquirrel.com**.

8.3 Using @font-face

If you have a font file in your project:

```
@font-face {  
  font-family: "MyFont";  
  src: url("MyFont.woff2") format("woff2");  
}
```

```
p {  
  font-family: "MyFont", sans-serif;  
}
```

9. ID vs Class in CSS

```
<p id="unique">This uses ID</p>
<p class="note">This uses Class</p>
<p class="note">Another element with same Class</p>
```

```
#unique {
  color: red;
}
```

```
.note {
  color: green;
}
```

- **ID** → used for one unique element.
 - **Class** → can be reused many times.
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10. Div in CSS

The `<div>` tag is a **container**. By default, it does nothing visually, but with CSS, we can make sections, boxes, or layouts.

Example: A colored rectangle

```
<div class="box"></div>
```

```
.box {
  width: 200px;
  height: 100px;
  background-color: lightblue;
  border: 2px solid blue;
}
```

👉 **Real-Life Example:** Think of `<div>` as a cardboard box. On its own, it's just empty. With CSS, you can paint it, resize it, and place it wherever you want.

Summary

- CSS styles HTML and makes web pages attractive.
- Three ways to use CSS: inline, internal, and external (best).

- CSS selects elements with tags, classes, or IDs.
- Text properties control color, fonts, size, and alignment.
- We can import and apply custom fonts.
- Divs are containers that can be shaped with CSS.